Linear Programming 1

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Introductory Example

A firm manufactures two products (only) at their Beeham factory. Both products require the use of the GG74 machine.

Product X requires 3 hours per unit at this machine

Product Y requires 2 hours per unit at this machine

In any week there are 144 hours of time available

Each unit of Product X gives a Contribution to Profit of £5, and

Each unit of Product Y gives a Contribution to Profit of £7.

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Linear Programming Model

Constraints

Production Constraint

- Product X requires 3 hours per unit at this machine
- Product Y requires 2 hours per unit at this machine
- In any week there are 144 hours of time available

$$3x + 2y \leq 144$$

Non-negativity

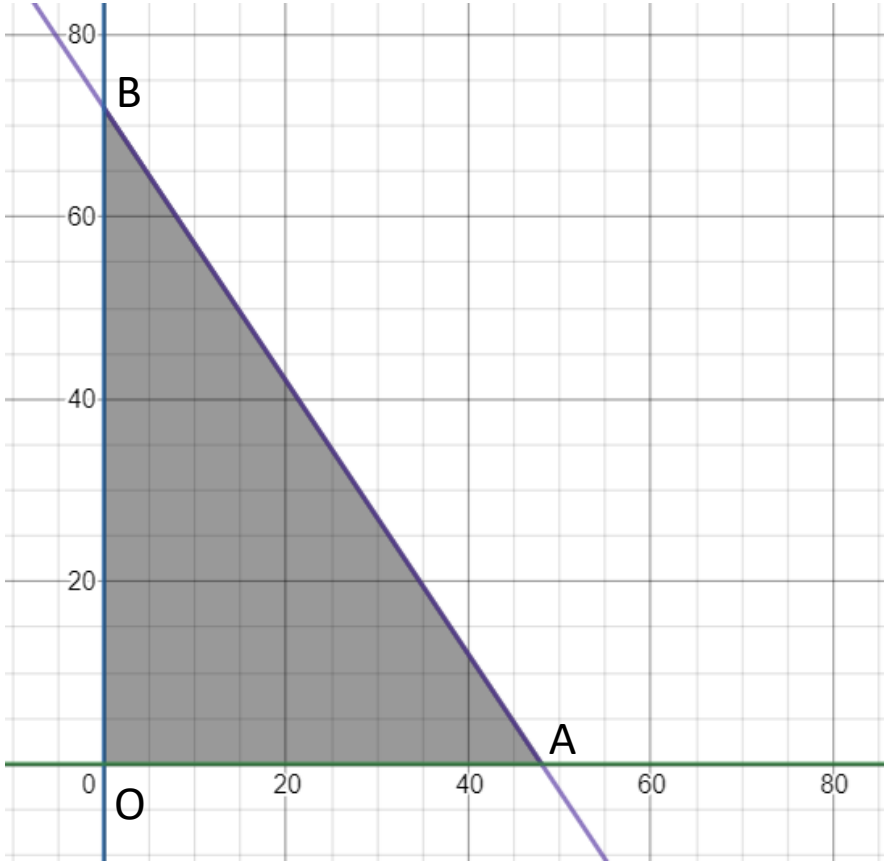
- It is not possible to manufacture a negative amount!

$$x \geq 0$$
$$y \geq 0$$

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Feasible Region



Any point within the feasible region is a solution to the system of inequalities.

$3x + 2y \leq 144$ (Constraints)

$x \geq 0$

$y \geq 0$

We can find the 3 points of intersection, O, A and B by solving the 3 pairs of simultaneous equations. E.g. for B…

$$3x + 2y = 144$$
$$x = 0$$

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Add an Objective

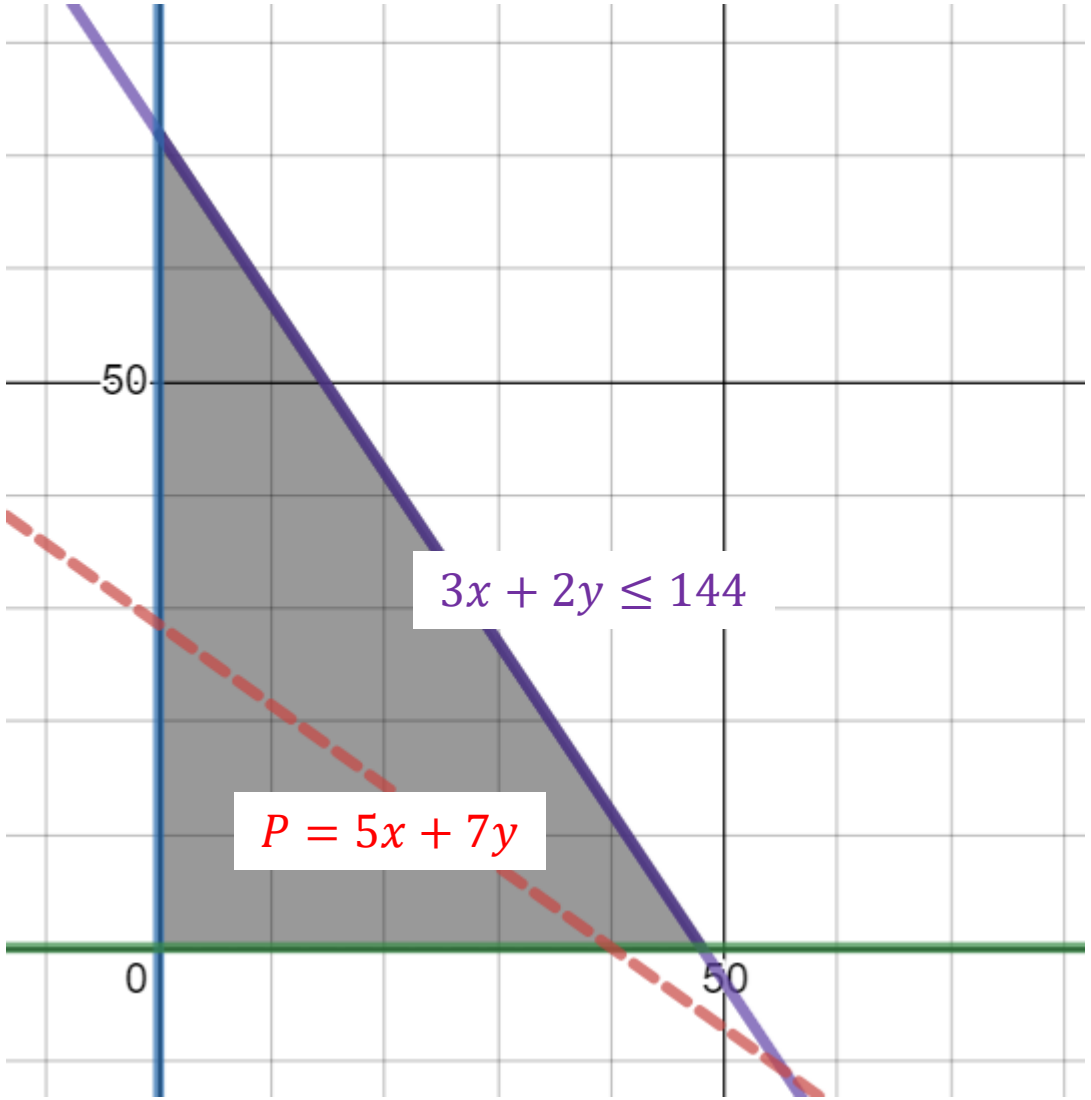
Maximise the profit

Each unit of Product X gives a Contribution to Profit of £5, and
Each unit of Product Y gives a Contribution to Profit of £7.

$$P = 5x + 7y$$

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Graphically $P = 5x + 7y$ is a straight line.

e.g. a Profit of $P = 200$ is represented by

$$5x + 7y = 200$$

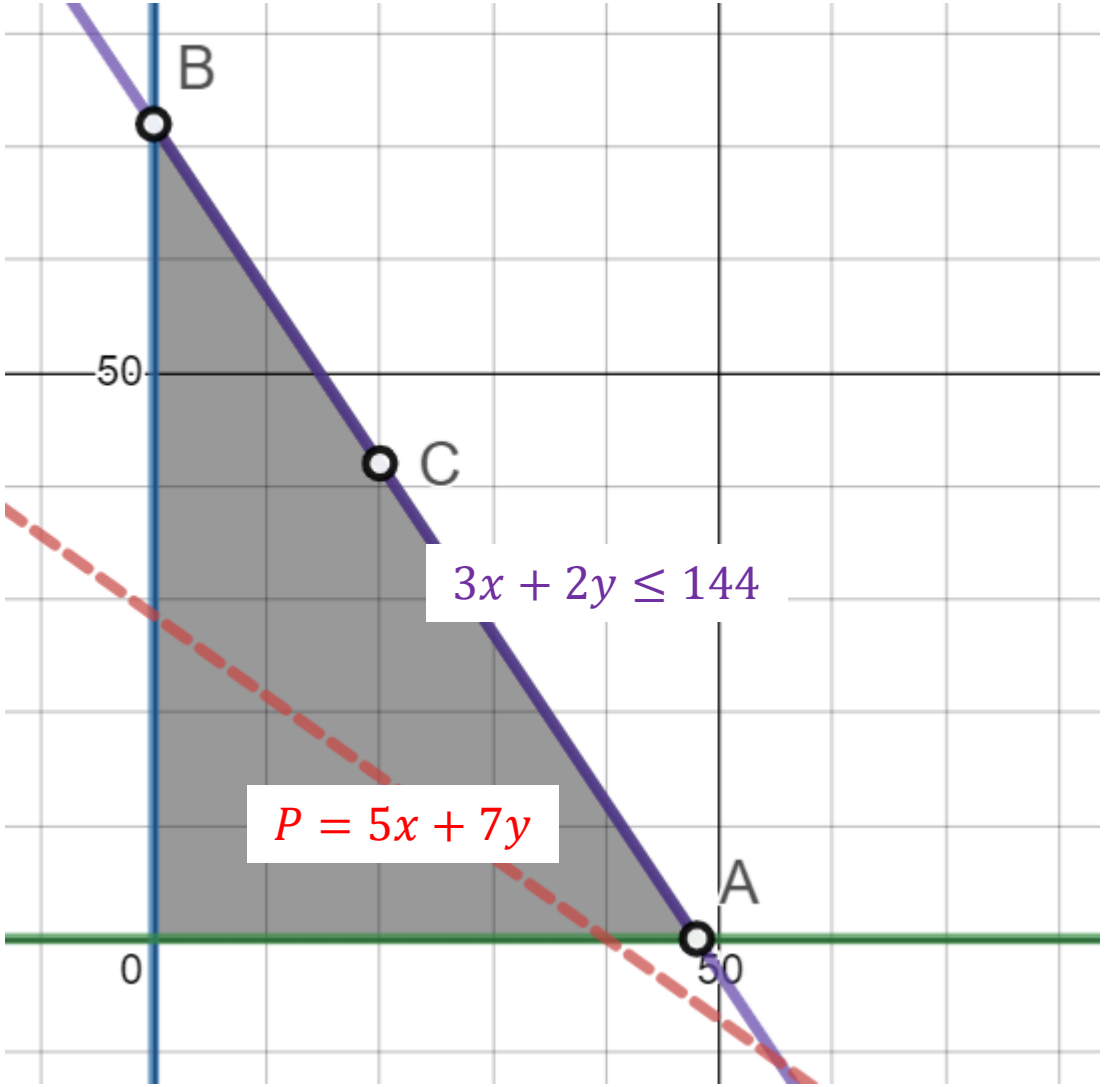
Choose a point (x, y) in the feasible region so as to maximise P

This is akin to moving the red profit line away from 0 as far as possible.

Interactive demo

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At A: $x = 48, y = 0, \quad P = 240$

At B: $x = 0, y = 72, \quad P = 504$

At C: $x = 20, y = 42, \quad P = 394$

Because the functions are Linear the **Profit** rises along AB from A to B.

Algebraically:

$$y = \frac{144 - 3x}{2}$$

$$P = 5x + 7\left(\frac{144 - 3x}{2}\right)$$

$$= -5.5x + 504$$

We are allowed values of $0 \leq x \leq 48$, hence the maximum P is at $x = 0$. Point B!

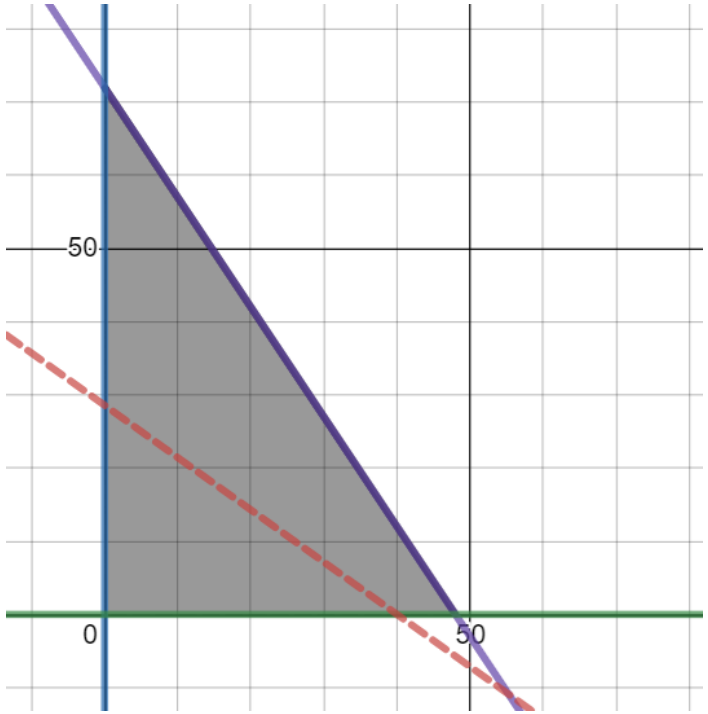
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Linear Programming Model

Our linear programming model can be written as follows

maximise $P = 5x + 7y$ (Objective)

subject to:



$3x + 2y \leq 144$ (Constraints)

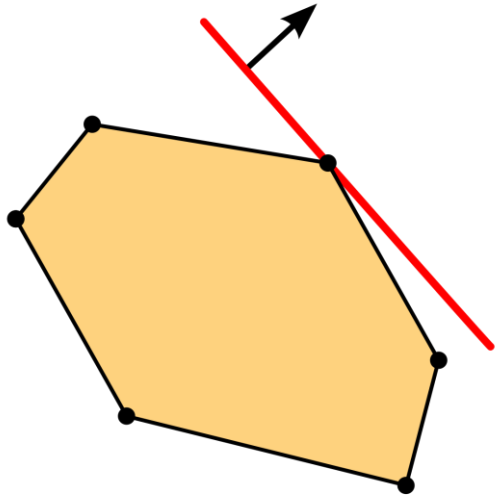
$x \geq 0$

$y \geq 0$

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Observations

- We had 1 constraint and 2 variables.
  - The optimal solution had one variable set to 0…
- The solution was at a vertex (corner of the feasible region)
- Intuitively we are moving the objective function (linear equation) over the feasible region until we exit.



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Is it a suitable solution?

QUESTION: Is this solution acceptable/workable for this Factory?

If not revisit the formulation.

For example, should the number of x produced be related to the number of y produced?

(Chairs x and Tables y) – Perhaps through experience the manager knows that tables will not sell on their own.

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Is it a suitable solution?

Introduce the constraint

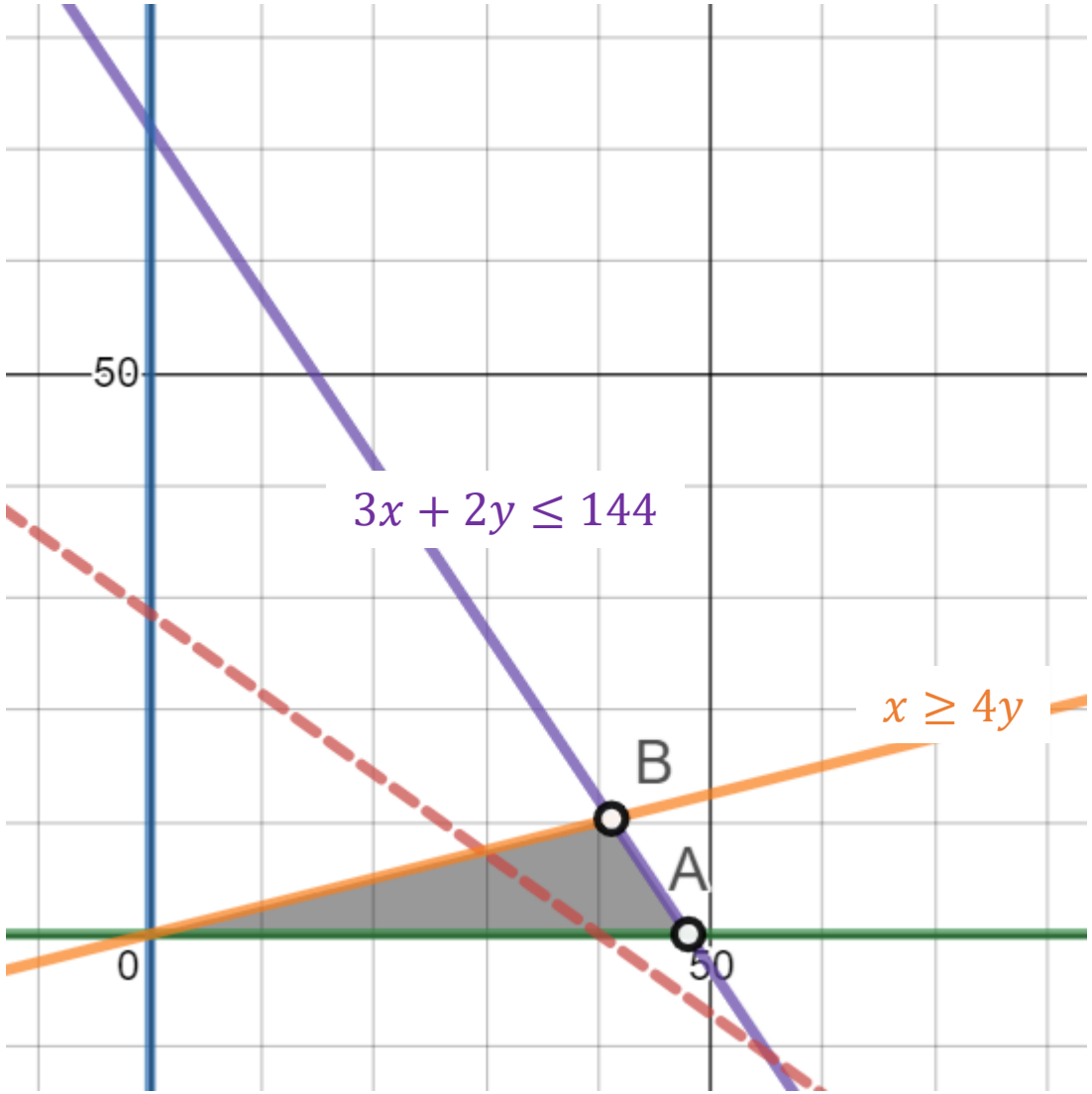
$$x \geq 4y$$

i.e. make at least 4 chairs for each table

Our profit P will now be maximised at B.

$$(x, y) = (41.14, 10.29)$$
$$P = £277.71$$

Note: We now have 2 binding constraints and the 2 variables are both non-zero (i.e. used)



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Linear Programming Model

Our linear programming model can be written as follows

maximise $P = 5x + 7y$

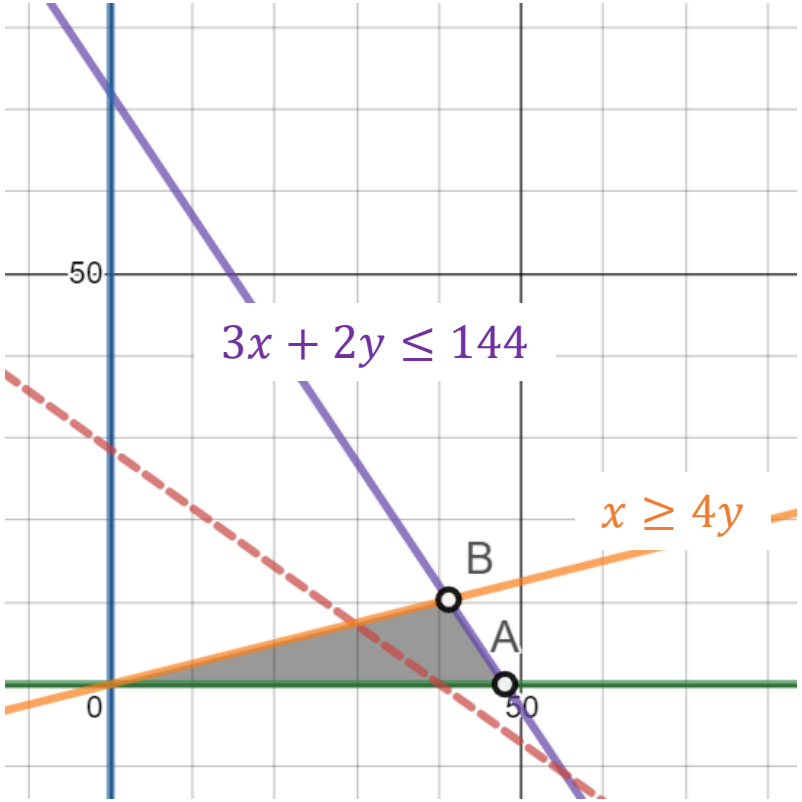
subject to:

$3x + 2y \leq 144$ (C1)

$x \geq 4y$ (C2)

$x \geq 0$

$y \geq 0$



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Binding Constraints

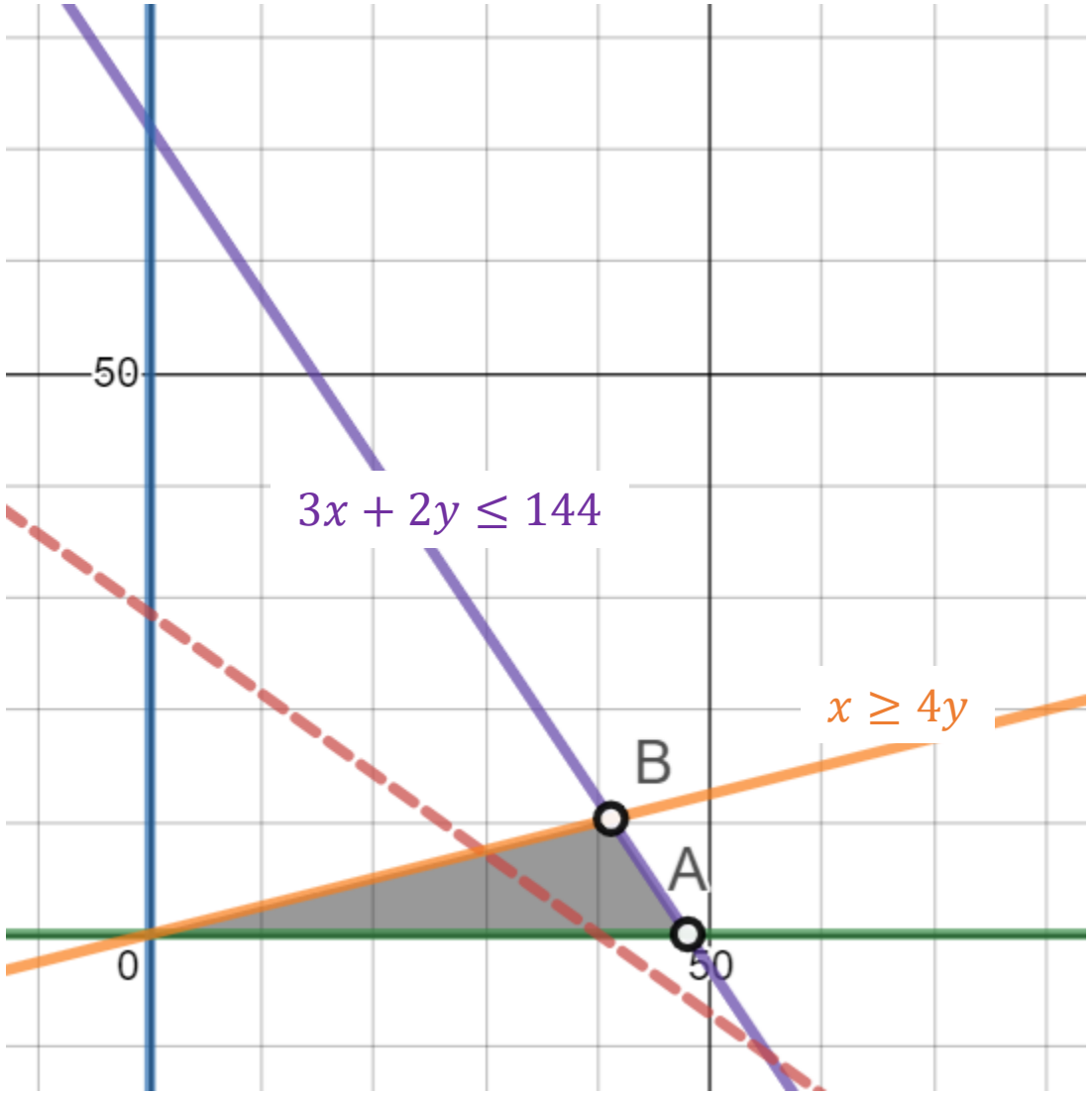
At **A**, $x = 48, y = 0$

$$3x + 2y = 144$$

Thus constraint (C1) is binding

$$48 \geq 4 \cdot 0$$

And constraint (C2) is not binding



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Binding Constraints

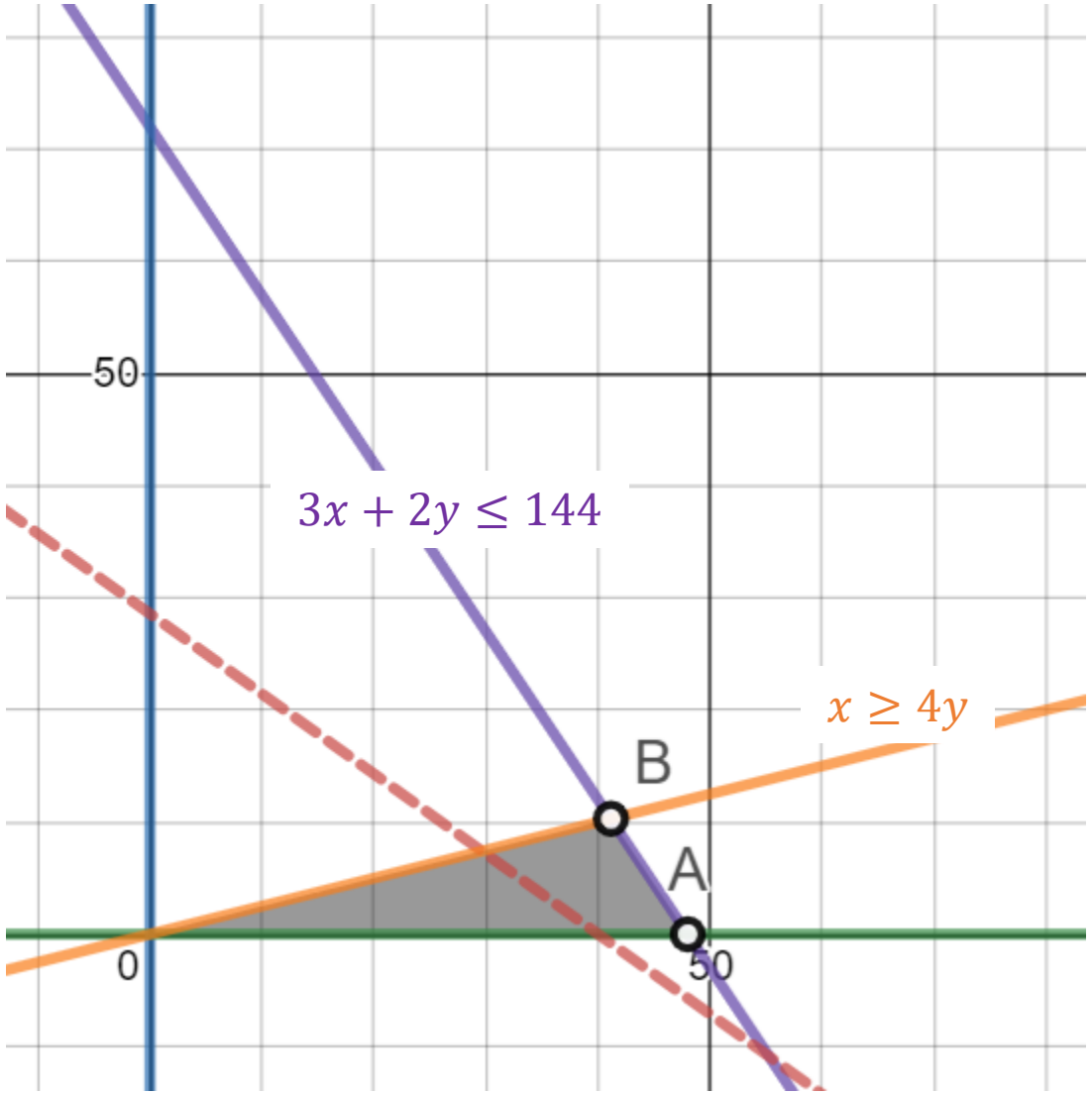
At **B**, $x = 41.14, y = 10.29$

$$3x + 2y = 144$$

Thus constraint (C1) is binding

$$41.14 \approx 4 \cdot 10.29$$

And constraint (C2) is also binding



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Infeasible

What about if the manager demands making 60 of product x a week?

$$x \geq 60$$

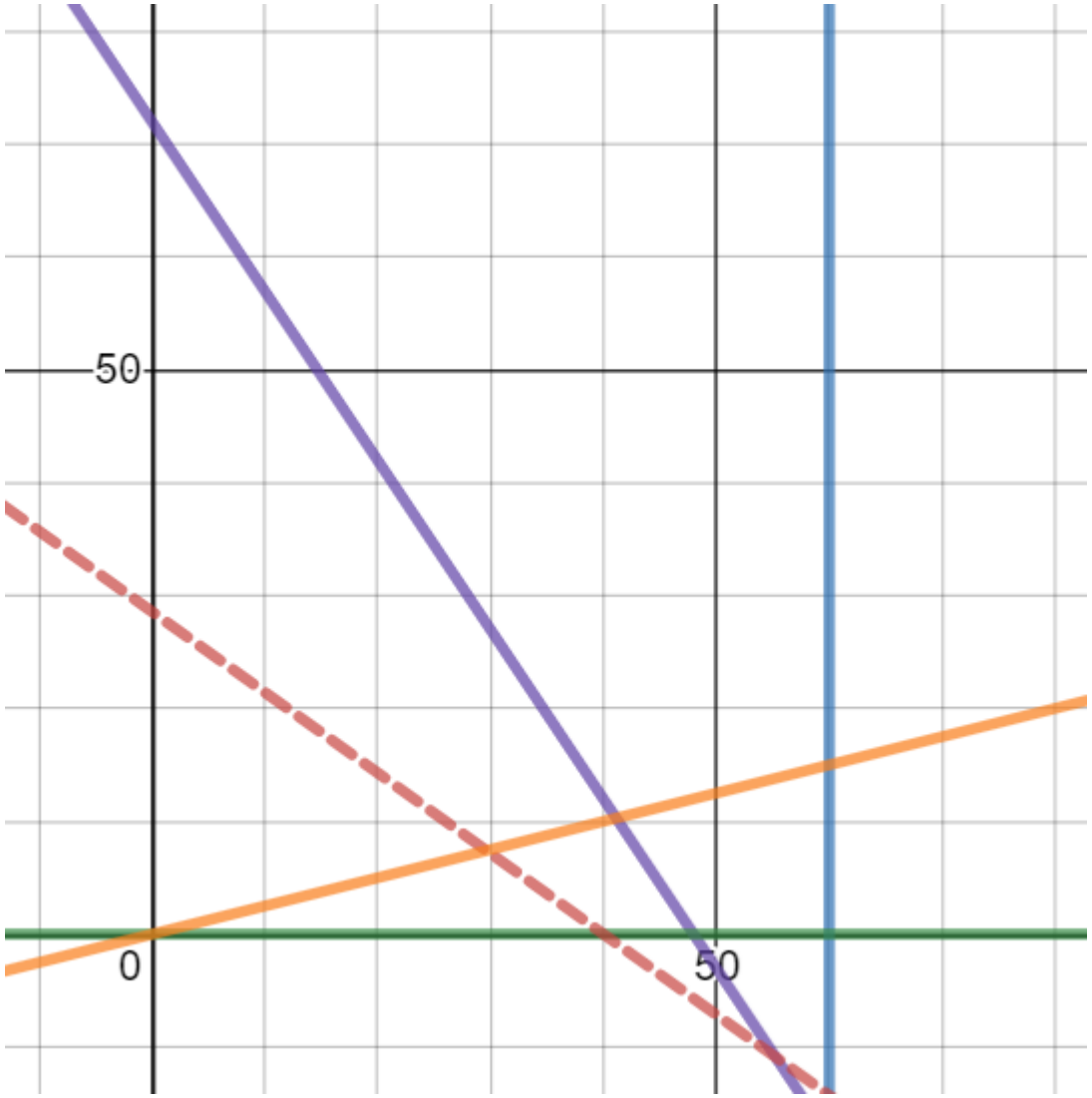
We can't simultaneously satisfy

$$3x + 2y \leq 144$$
$$x \geq 60$$

i.e. $3 \cdot 60 = 180 > 144$

This is known as an infeasible problem, there is no feasible region!

Constraints are inconsistent



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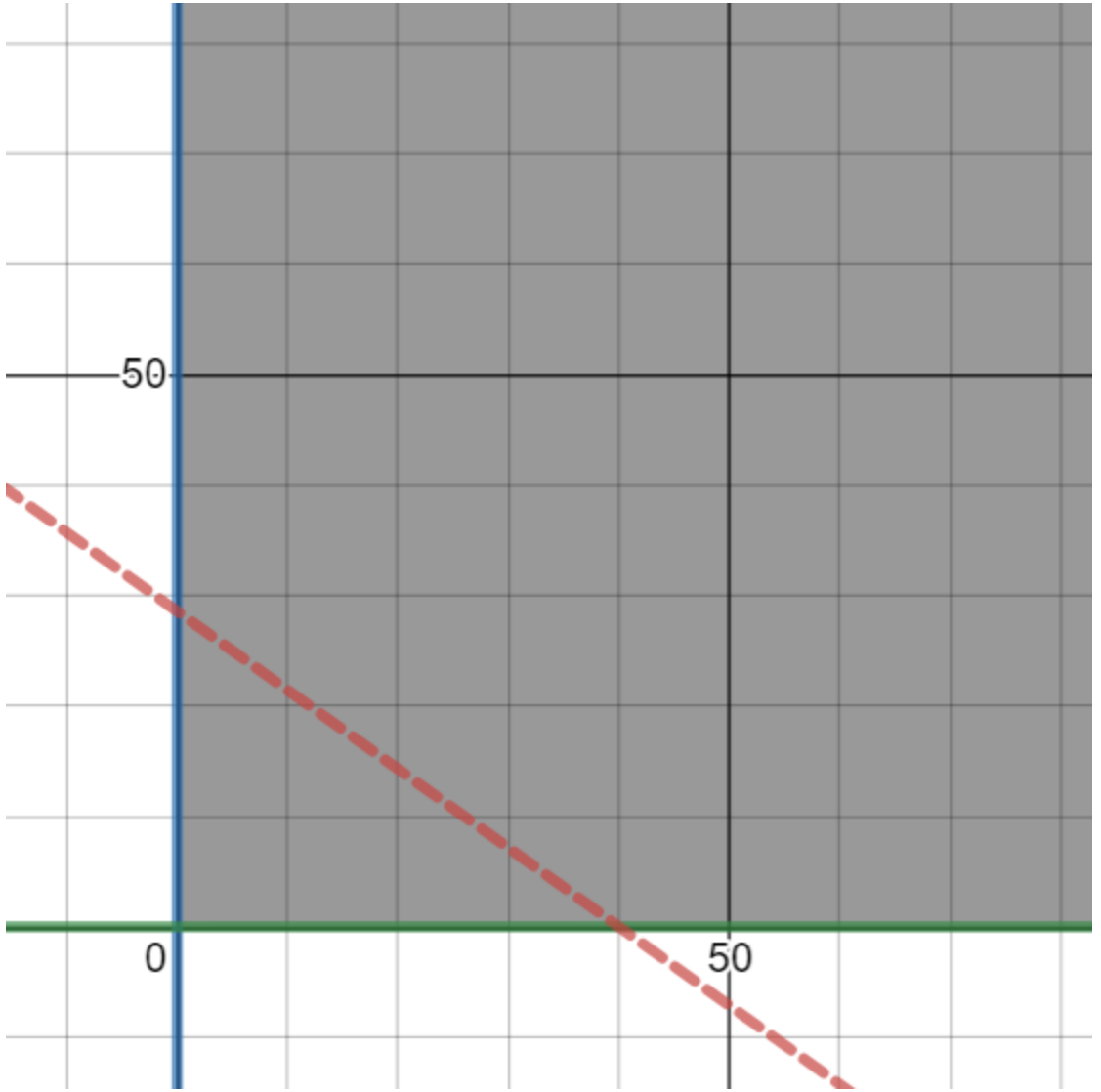
Unbounded

What about if we remove both binding constraints

$$3x + 2y \leq 144$$
$$x \geq 4y$$

Our profit line $P = 5x + 7y$ is now free to move to an unlimited profit (grey feasible region).

This is known as an unbounded problem, there is no restriction on the objective



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Infinite Solutions

If our objective is parallel

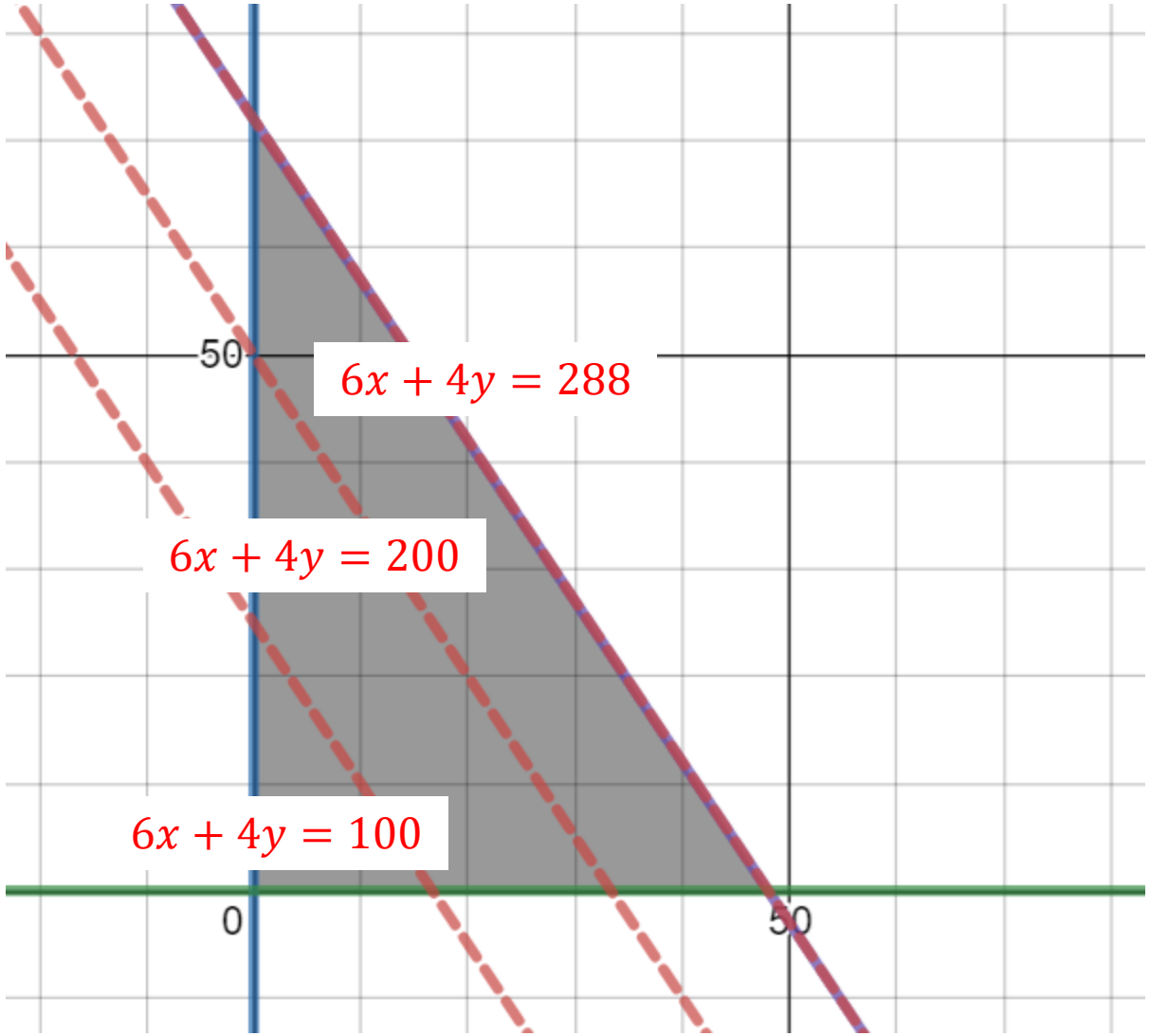
maximise $P = 6x + 4y$

subject to:

$$3x + 2y \leq 144$$
$$x \geq 0$$
$$y \geq 0$$

Optimal solution anywhere on the line $3x + 2y = 144$ from (0, 72) to (48, 0). Thus $P = 288$

Note that it still has an optimal solution at a vertex!



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Optimality

An optimal solution always lies on the boundary of the feasible region: at a vertex, or along a whole edge (infinitely many solutions).

If an optimal solution exists, one is always at a vertex. The vertices are known as Basic Feasible Solutions.

A Basic Feasible Solution (BFS) is the intersection of a number of boundaries.

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Graphical illustration of 3 variables

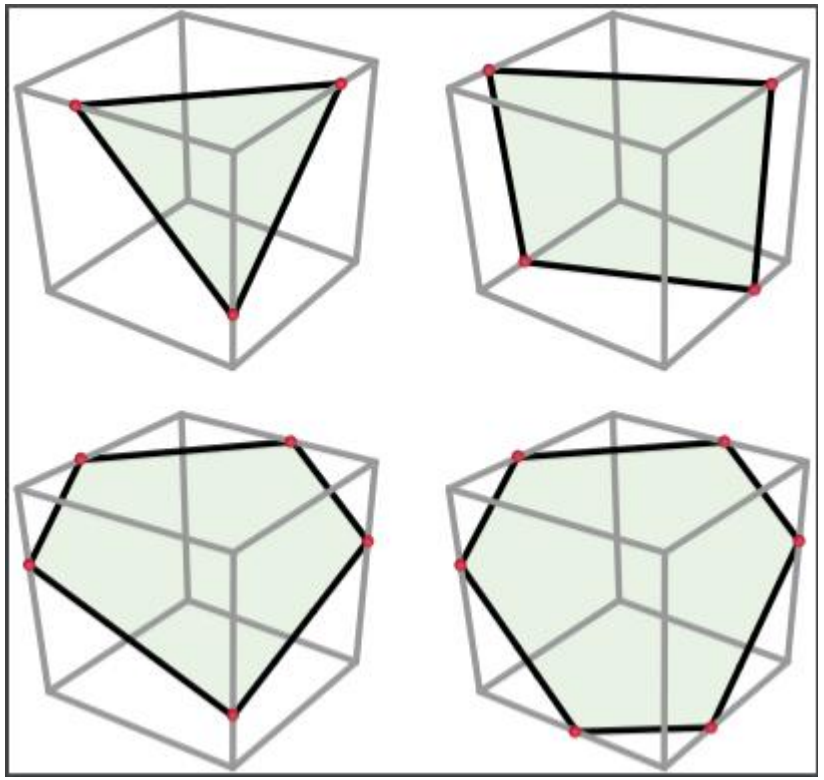
Intersection of different planes with a cube

e.g.

minimise $P = x + 3y + z$

subject to:

$$x \leq 1$$
$$y \leq 1$$
$$z \leq 1$$
$$x \geq 0$$
$$y \geq 0$$
$$z \geq 0$$



Each corner of the cube is a BFS.

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Canonical Form (Matrices)

Our linear programming model

maximise $5x + 7y$

subject to:

$$3x + 2y \leq 144$$
$$x \geq 4y$$
$$x \geq 0$$
$$y \geq 0$$

maximise $5x + 7y$

subject to:

$$3x + 2y \leq 144$$
$$-x + 4y \leq 0$$

and

$$x \geq 0$$
$$y \geq 0$$

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} x \\ y \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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Canonical Form (Matrices)

Let

$$\boldsymbol{c} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \boldsymbol{A} = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

→

Canonical Form

maximise $\boldsymbol{c^T x}$

subject to: $\boldsymbol{Ax \leq b}$

and $\boldsymbol{x \geq 0}$

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In general

Let

$$\boldsymbol{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad \boldsymbol{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Canonical Form

maximise $\boldsymbol{c^T x}$
subject to: $\boldsymbol{Ax \leq b}$
and $\boldsymbol{x \geq 0}$

e.g

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \\ 100 \end{bmatrix}$$

and

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

What are m and n for the example?

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Maximum and minimum equivalence

maximise $\boldsymbol{c^T x}$
subject to: $\boldsymbol{Ax \leq b}$
and $\boldsymbol{x \geq 0}$

Is equivalent to…

minimise $\boldsymbol{-c^T x}$
subject to: $\boldsymbol{Ax \leq b}$
and $\boldsymbol{x \geq 0}$

Additional homework, why?

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Solution via Python

maximise $\begin{bmatrix} 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

subject to:

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 144 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} x \\ y \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We are maximising.

The Python function `linprog` minimises!

Notice that in the Python code we set c = [-5, -7]

```python
from scipy.optimize import linprog

# cost function (row vector) - to minimise we need the negatives
c = [-5, -7]

# linear inequalities
# (no. rows in A = no. rows in b)
A_ub = [[3, 2],
        [-1, 4]]
b_ub = [144, 0]

#  call the linprog function
results = linprog(c, A_ub, b_ub)

print(results.message)
# if results are a success then print out the optimal solution
if results.success:
    for k in range(len(results.x)):
        print(f"x[{k+1}] = {results.x[k]:.2f}")
    print(f"fval = {results.fun:.2f}")
```

[scipy.optimize.linprog — SciPy documentation]

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Solution via Python

We will use a friendlier way to express this next week, however it is important to:

- get to understand the canonical form
- think about this in terms of Linear Algebra

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Tutorial

- Solve a linear program with Python
- Solve a linear program with Excel
- Model a simple LP problem and solve
- Model a more complicated LP, we will start the beginning of the next lecture with a discussion about this.

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